DONGBA SCRIPTS LEARNING APP

An AR-based learning experience for exploring Naxi pictographic writing and cultural heritage.

Overview

Dongba Script Learning APP is an interactive learning app designed to make Naxi cultural heritage more accessible through augmented reality (AR). By scanning Dongba characters and assembling their separated 3D components, users can discover how these pictographic symbols represent real-world objects and actions. Animated character reconstructions and contextual information transform traditional script learning into an engaging, hands-on experience, connecting visual exploration with cultural understanding.

Interaction Design | UI/UX Design | AR Experience Design
Tools: Figma, Unity, Photoshop

INSPIRATION

The traditional Dongba Script is experiencing a significant decline, with fewer than 100 fluent speakers left today. Its preservation has become an urgent cultural imperative, as the script embodies profound historical and linguistic value, serving as a vital cornerstone for reinforcing Naxi ethnic identity and safeguarding cultural heritage.

This inspired me to think about what we can do to **make this endangered language accessible** to younger generations.

Language History

Dongba Script is one of the world's few living pictographic languages, carrying thousands of years of Naxi history, rituals, and cosmology.

Facing Challenges

The complex characters and lack of structured learning paths make it difficult for young people to start learning Dongba Script.



Language Education

Modern education in Naxi communities focuses on Mandarin, leaving little room for Dongba Script learning in daily classrooms.

Cultural Erosion

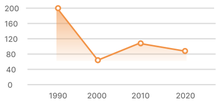
Most young Naxi people cannot read or understand Dongba Script, risking the permanent loss of their linguistic heritage.

BACKGROUND

The decline of Dongba Script speakers has become an undeniable reality, leading to the risk of losing a living pictographic writing system forever.

Speaker Decline

number of speakers

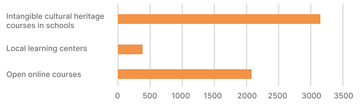


Fewer than 100 speakers today

Only 0.03% of the 324,000 Naxi population can read and write proficiently

Dongba Revival

Growing Engagement with Dongba Script Learning



Over 5,000 learners

through online and offline education programs

Conclusion

It is essential to let young people understand the importance of Dongba Script to **prevent its extinction**. Losing this writing system is also losing a part of Naxi culture. With native speakers declining, **supporting new learners** is an effort we must make.

DESK RESEARCH

The decline of Dongba Script learners stems from multiple interconnected barriers. High complexity, outdated teaching methods, an aging teaching force, and a lack of real-world usage together discourage young people from engaging with the language.

High Learning Barrier

The complex pictographic system and thousands of unique characters create a steep learning curve for new learners.

Elderly Speakers

Most fluent speakers and teachers are elderly, with few young people to pass on the language to new generations.



Outdated Teaching Methods

Most learning relies only on static textbooks, lacking diverse or interactive methods to engage young learners.

Lack of Daily Use

There are almost no daily contexts to use Dongba Script, making it hard for learners to practice and retain knowledge.

Internet - Case Study

Old Norse (Northern Europe)

Old Norse is no longer spoken natively, surviving only in ancient sagas and myths. Its complex grammar and runic writing system make it extremely hard to learn, and with no living speakers, it can only be studied through static texts.
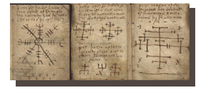




Aka Language (South Asia)

The critically endangered Aka language has fewer than 2,000 fluent speakers left, most of whom are elderly. With no young inheritors to pass it down, it faces imminent extinction.

Rapa Nui Language (Oceania)

On Easter Island, most young people now speak Spanish as their daily language. Rapa Nui has lost its place in homes, schools, and public life, with almost no real-world usage scenarios left to keep the language alive.



Summary

Language endangerment is a widespread global issue. Across different regions, languages are declining due to common threats: an aging speaker base, lack of daily use, and high barriers to learning. Preserving these languages is not just about saving words, but about protecting irreplaceable cultural heritage.

PROBLEM STATEMENT

Ancient Dongba Script Is Beautiful but Hard to Access

As a precious pictographic cultural heritage of the Naxi people, Dongba script carries local stories and traditional lifestyles. However, most tourists, students and museum visitors can hardly understand abstract ancient characters through static posters, paper guidebooks and traditional exhibition forms. The monotonous display method raises the learning threshold and prevents young users from further exploring traditional culture.

FIELD RESEARCH

This field research mainly targets three core user groups: cultural tourists, school students and museum visitors. The research was conducted in Dongba cultural exhibition halls, ancient town scenic spots and intangible cultural experience stores to analyze user behaviours and existing pain points during cultural learning.

Field Observation



Passive Receiving of Cultural Information

Most tourists and student visitors only take photos of murals, wooden carvings and exhibition boards as simple check-in behaviours. They have no effective ways to interpret the meaning and cultural origin of Dongba characters.



High Learning Barrier for Abstract Pictographs

Dongba script relies heavily on pictographic modelling. Beginners cannot recognise character structures, evolution logic and corresponding real objects merely through static pictures.



Limited Explanation from Traditional Carriers

Paper textbooks and museum captions can only provide text descriptions. They fail to demonstrate the dynamic connection between ancient characters and natural objects, animals and farming scenes.

Scene Classification



Museum Static Exhibition

Fixed-angle display, lack of interactive content



Ancient Town Scenic Billboard

Users cannot obtain extended cultural information instantly



Offline Dongba Writing Course

High cost, limited opening time and venues

FIELD RESEARCH

The AEIOU analysis aims to explore the problems of conventional Dongba script learning and support the design of Dongba AR learning application.

AEIOU

A ctivities

Visit Cultural Exhibitions View Dongba Murals Participate in Cultural Experience Courses Take Photos for Collection Learn and Memorize Ancient Characters

E nvironments

Museum Exhibition Area Ancient Town Scenic Area Intangible Cultural Heritage Workshop School Cultural Classroom Online Learning Platform

I nteractions

Browse Static Exhibition Boards Read Text Explanations Communicate with Instructors Take Pictures of Exhibits Insufficient Interactive Ways between Users and Ancient Characters

O bjects

Dongba Paper Stone Stele Wooden Totem Mural Carving Printed Textbook AR Identification Model and Pictographic Character Resources

U sers

Cultural Tourists Primary and Secondary School Students College Students Museum Visitors Traditional Culture Enthusiasts






Insights

Single and outdated cultural display methods

Traditional exhibitions and textbooks rely excessively on static text and pictures. It is difficult for tourists and student users to establish an intuitive connection between pictographic characters and real objects.

High threshold for independent learning

Most beginners cannot master the evolution logic and structural composition of Dongba scripts without professional guidance, which easily causes learning frustration and abandonment.

Users have strong willingness to explore traditional culture

Tourists and young students are curious about exotic ancient characters. They prefer immersive, visualized and interactive learning methods rather than rigid textbook education.

PRIMARY RESEARCH

After understanding the pain points in current Dongba cultural learning, I aim to further explore the perspectives of tourists, students and museum visitors on Dongba script education and interactive heritage experience.

Questionnaire

The data were collected through online surveys, on-site distribution in scenic spots and museums, and interviews with young learners.

Question

Have you ever encountered Dongba script in travel, museum visits or daily life?

Do you find it difficult to understand abstract pictographic characters like Dongba script?

What ways have you tried to learn or understand Dongba script?

Would you be interested in learning Dongba script through mobile apps? Have you ever used AR technology in a museum or exhibition?

What do you think is the most interesting part of Dongba script?

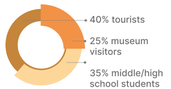
Would you prefer short, gamified learning activities or long, textbook-style courses?

Do you think learning Dongba script can help you better understand Naxi culture?

Have you ever felt frustrated or bored when learning traditional culture through static materials?

If there was an AR app that lets you scan Dongba characters and interact with them, would you use it while traveling or visiting museums?

Data



Basic information of participants

It is concluded that tourists and students have strong curiosity about Dongba culture, but are blocked by the high learning threshold and lack of interactive forms. There is a clear market demand for a mobile AR app that combines cultural education with gamified interaction, especially for use during travel and museum visits.

Interview

To better understand the needs of different users for Dongba culture learning, I conducted semi-structured interviews with three core groups: cultural tourists, student learners and museum visitors.

Cultural Tourist

When you travel to Lijiang, how do you usually learn about Dongba script? Do you feel interested or confused when you see these characters on signboards?

I take photos of the Dongba murals and signboards as check-in photos. But sometimes I have no idea what the characters mean, and there's no easy way to find out.

Student Learner

Have you ever learned about Dongba culture at school or online? What do you think is the biggest difficulty?

I've seen Dongba script in textbooks, but it's too abstract. I can't connect the characters with real objects. It's hard to remember their meanings without vivid examples.

Museum Visitor

When you visit a Dongba culture exhibition, how do you usually interact with the exhibits? Do you have any expectations for new interactive forms?

Some museum exhibits are too technical and hard to understand, especially for my kid. He gets bored quickly when just looking at static pictures. If we could see the characters come alive with animations, it would make learning more interesting.

Findings

Most tourists lack convenient ways to learn the meaning and cultural stories behind the characters.

Students and young learners face a high learning threshold when trying to understand abstract pictographic characters like Dongba script.

The majority of young users already use mobile phones and apps for leisure.

Persona



Basic Info

- Age: 16
- Gender: Male
- Occupation: High School Student
- Location: Lijiang, Yunnan
- Ethnicity: Naxi

Needs & Pain Points

- Wants to systematically learn Dongba script and pass on the culture of his ethnic group.
- Finds traditional textbooks boring and hard to memorize.
- Hopes to have an app with structured learning paths, pronunciation teaching, and practice functions.



Needs & Pain Points

- Needs a portable tool to quickly check Dongba script meanings and cultural allusions when leading tours.
- Wants AR functions to show tourists vivid demonstrations of characters on murals and signboards.
- Hopes the app can help make cultural experiences more professional and fun.





Basic Info

- Age: 32
- Gender: Female
- Occupation: Tour Guide / Cultural Tourism Practitioner
- Location: Lijiang, Yunnan



Basic Info

- Type: Cultural Institution & Intangible Heritage Protection Unit
- Location: Lijiang, Yunnan
- Core Role: Dongba culture preservation, education and promotion

Needs & Goals

- Needs digital tools to help protect and spread Dongba culture.
- Hopes the app can be an official cultural promotion carrier for research trips and cultural tourism projects.
- Wants to expand the influence of intangible heritage and realize mutual empowerment between cultural inheritance and tourism development.



IDEATION

Design Goal

Help tourists, students and museum visitors understand Dongba script through interactive AR experiences, and inherit and spread Naxi culture in an accessible way.

Different Ways

	Advantage	Limitation
Virtual Reality	· Immersive experience · Rich virtual environments	· Specialized equipment · Limited accessibility
Interactive Installation	· Shared public experience · Large-scale interaction	· High production cost · Fixed location
AR Learning App	· Accessible on smartphones · Interactive 3D learning	· Dependent on device performance · Requires clear interaction guidance

Final Choice - AR Learning App

Accessible Learning
Smartphone-based AR lowers the barrier to exploring Dongba script in museums, classrooms, and everyday settings.

Active Participation
Users reconstruct characters by dragging 3D components instead of passively viewing information.

Cultural Understanding
Character animations and contextual explanations connect visual forms with their meanings and cultural significance.

STRATEGY

Interaction



AR scanning activates the learning experience directly from printed Dongba characters.

Users drag and drop separated 3D components into the correct positions, turning character learning into an active, hands-on task.

3D Visualization



Each character is decomposed into individual 3D components, allowing users to examine its structure from a new perspective.

Once correctly assembled, the character comes to life through animation, making the interaction more intuitive and memorable.

Gamified Learning



A task-based learning flow encourages users to learn by doing, with successful assembly providing immediate visual feedback.

The app's game-like structure can support repeated practice and gradual learning, making cultural exploration more approachable.

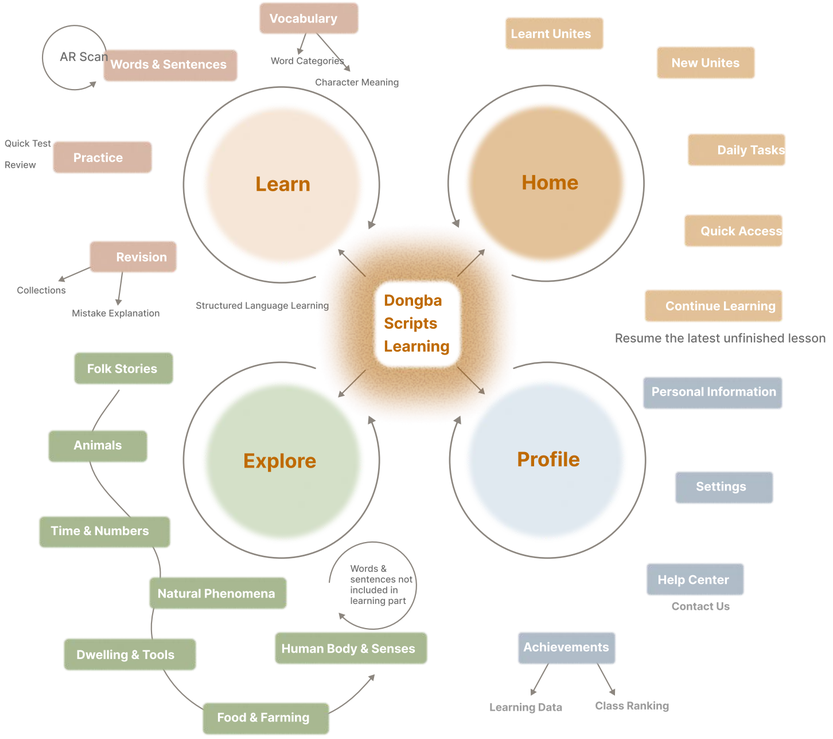
Cultural Information



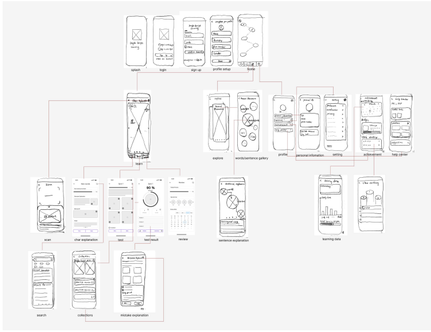
After the character is completed, an information panel presents its meaning and cultural context.

Combining interaction with concise explanations helps connect the digital experience to the living heritage of the Naxi people.

INFORMATION ARCHITECTURE



PAPER PROTOTYPE



Home
The Home interface provides quick access to the app's main features and learning content. Users can discover new learning units, continue previous lessons, and access daily learning activities.

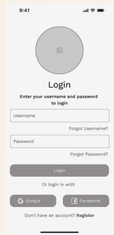
Learn
The Learn interface offers structured Dongba script learning through vocabulary, character explanations, and interactive exercises. Users can scan characters with AR, explore their meanings, take tests, and review mistakes to reinforce their learning.

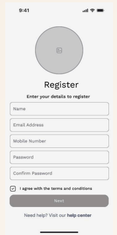
Explore
The Explore interface allows users to discover Naxi culture through different themes, including folk stories, animals, natural phenomena, daily life, and traditional practices. By exploring these topics, users can connect Dongba script with the cultural knowledge and everyday experiences behind it.

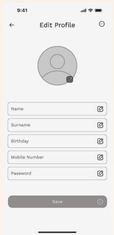
Profile
The Profile interface brings together users' personal information, learning progress, and achievements. Users can track their learning data, review their achievements and rankings, manage app settings, and access the Help Center.

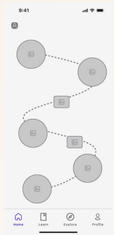
LOW FIDELITY

Splash


Login


Register


Profile Setup


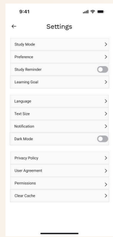
Home


Learn

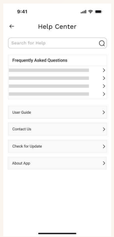

Explore


Profile

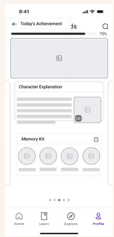

Personal Information


Settings


Achievement

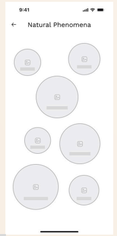

Help Center


Vocabulary

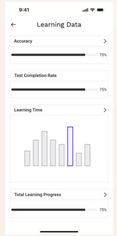

Words Explanation


Test


Review


Words Gallery


Sentence Explanation


Learning Data


Class Ranking


Scan


Test Result


Mistake Explanation


Collections


HIGH FIDELITY

Typo System

Logo

The Dongba Script Learning logo uses the Dune font to create a distinctive visual identity. Its expressive letterforms complement the app's cultural theme, while the overall design aims to balance a contemporary aesthetic with the visual character of traditional Dongba script.

Typography

Work Sans is used for interface titles and body text, providing a clean, modern, and highly readable foundation for the app. Roboto is applied to subtitles and secondary labels, creating a clear visual hierarchy and ensuring consistency across different interface elements.

Interface Titles and Body Text

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
0 1 2 3 4 5 6 7 8 9
! ? # % & @

Subtitles and Secondary Labels

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
0 1 2 3 4 5 6 7 8 9
! ? # % & @

Color

The interface adopts a warm, earthy palette inspired by traditional paper and the visual character of Dongba culture. Cream and beige establish a calm, welcoming background, while terracotta orange highlights key actions and navigation elements. Dark brown provides contrast for text and icons, creating a cohesive visual language that connects cultural warmth with a contemporary digital experience.





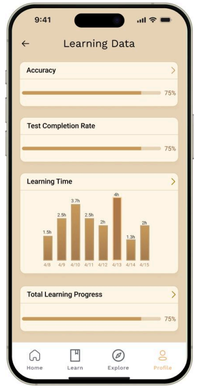


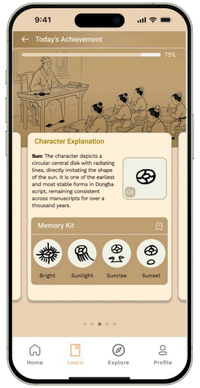


An AR-based learning experience for exploring Naxi pictographic writing and cultural heritage.

The logo symbolizes the connection between learning and cultural heritage. Two semicircles represent hands holding an open book, from which a Dongba pictograph meaning "tree" emerges. This visual metaphor reflects the idea of bringing traditional script to life through learning and exploration. By combining the book, hands, and pictographic symbol into a single graphic, the logo conveys the app's mission to make Dongba script more accessible while preserving the cultural knowledge embedded in its visual forms.







AR

Process

1 Scan and Detect

Scan a Dongba character on the page to activate the AR experience.



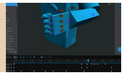


Assemble the Character 2

Drag and drop the separated 3D components into the correct positions to reconstruct the character.

3 Trigger the Animation

Once all components are correctly placed, the complete 3D character comes to life through animation.





Reveal the Meaning 4

An information panel reveals the character's meaning and cultural context.

Design Breakdown

Page Design



The learning pages introduce Dongba characters through clear visual layouts and illustrated examples. Each page presents a character alongside its meaning and related cultural context, helping users understand.

3D Models



Dongba characters are reconstructed as interactive 3D models composed of separate components. Users drag and place these components to complete the character, triggering an animation when the assembly is successful.

Info Panel



The information panel provides explanations of each character's meaning, origin, and cultural significance. By combining concise text with traditional illustrations, it connects the interactive learning experience with the cultural knowledge behind Dongba script.

Technical Implementation

I developed two custom C# scripts in Unity, DragItem and DropZone, to implement the character assembly interaction

DragItem

DragItem manages object movement and tracks the active drop zone,

Drop-Zone Detection
Tracks the active drop zone as the component moves, preparing the system to validate where the user releases it.



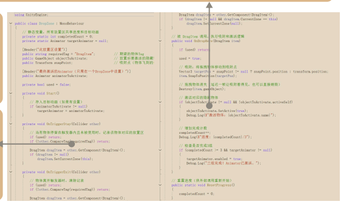
Drop Handling
When the user releases a component, the script passes it to the active drop zone for placement handling.

Placement Validation
Checks whether the incoming object has the required tag before processing the placement, filtering out unrelated objects.

DropZone

DropZone validates object placement, snaps components into position, and activates the corresponding visual elements.

Visual Activation & Progress
Activates the corresponding visual object and increments the completion counter after a component is accepted.



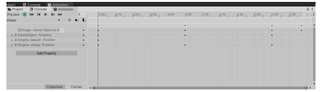
Animation





A stone hammer falls, gradually reducing the stone's size to illustrate the action of chiseling and bring the pictographic character to life.





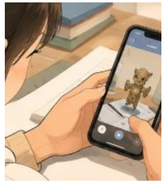
A herding stick swings down as the sheep moves to the right, visually demonstrating the movement represented by the character.





An arrow flies forward from the bow, transforming the static pictograph into a dynamic scene that illustrates the action of shooting.

Hypothetical User Testing

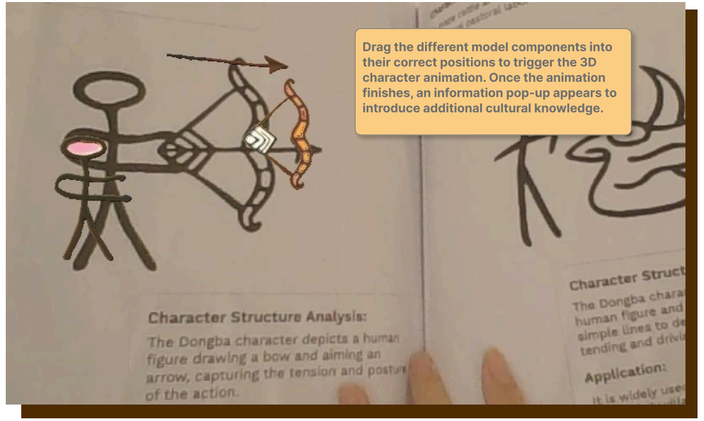


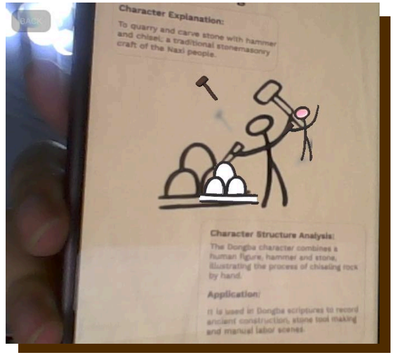
"The 3D characters make learning Dongba script much more engaging. I especially like how the AR experience brings the symbols to life and lets me interact with them."

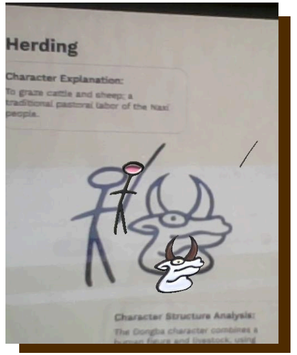
"The character assembly is an interesting way to learn, but I would like more explanations of what each component means and how it relates to the complete symbol."



FINAL DESIGN







REFLECTION

Overview

The core goal of this Dongba Script AR App is to help tourists, students and museum visitors learn pictographic Dongba characters via interactive AR gameplay. Users scan real-world Dongba characters on murals, signboards or exhibits to load split 3D character parts; assembling these fragments into complete characters by drag-and-drop triggers origin animation and detailed pop-up information about culture and pronunciation.

Challenge



· 3D segmented model production for each individual Dongba character

· Combine real-scanned physical Dongba script with split AR 3D models

· Design drag-and-drop assembly rules and attractive animated feedback to retain users

Future Outlook



· Cooperate with Lijiang Dongba Culture Museum and local primary & secondary schools

· Add more folk stories and Naxi daily-life content into pop-up introduction

· Work with intangible heritage inheritors to launch physical Dongba paper peripheral products

Summary

This project helps me master UI layout rules and upgrade my skills of making segmented 3D models on Blender. During the design cycle, I not only polish interactive AR design ability but also build systematic cognition of cultural-oriented product development. Moving forward, I will keep optimizing AR interaction logic and explore new interactive forms for intangible cultural heritage dissemination.